

CHINMAY GOYAL

Senior Year Undergraduate — Computer Science and Engineering — IIT Kanpur

☎ (+91)789-155-5552 ✉ chinmayg@(cse.)iitk.ac.in | cg2chinmay@gmail.com 📍 chinmaygoyal in chinmaygoyal

Education

Year	Degree	Institute	CPI/Percent
2018 - Present	B.Tech., Computer Science and Engineering (Grad. 2022)	Indian Institute of Technology Kanpur	9.8/10.0
2018	Class XII (CBSE - AISSEE)	Rukmani Birla Modern High School, Jaipur	95.2%
2016	Class X (CBSE - AISSE)	Rukmani Birla Modern High School, Jaipur	10.0/10.0

Achievements

- Secured a place among top 35 teams at the **ICPC 2020 Asia Amritapuri Regional Contest** across top 750+ selected teams
- Received **Academic Excellence Award** for the years 2018-19 and 2019-20 at IIT Kanpur
- Secured an All India Rank **166**, among 0.23 million candidates in **JEE Advanced** 2018, organized by IIT Kanpur
- Achieved an All India Rank **817**, among 1.2 million candidates in **JEE Main** 2018, conducted by CBSE
- Attained All India Rank **153** and selected for **Kishore Vaigyanik Protsahan Yojana**(KVPY) fellowship in 2016-17
- Secured **National Talent Search Examination, 2016** (NTSE) scholarship and **State Rank 1** (Stage-1)
- Among **National Top 1%** in **National Standard Examination in Physics**, 2017 organised by IAPT
- Secured a place among top 650 participants in Google Kisck Start, 2021
- Achieved highest rating of **1842** on Codeforces with active participation in contests.

Internship Experience

TOWER RESEARCH CAPITAL — Project: **An Aggressive Model for Taking** [May '21 - July '21]

- Objective: Enforce strategies to generate an **aggressive trading model** to execute profitable trades based on market behaviour
- Constructed model to predict the market behaviour of a product using **classification** and **regression** techniques
- Compared performance of different models through variations in **Precision-Recall Curves**, **F Scores** and **Profit-Loss Ratio**
- Achieved an increment of profit by **150%** in comparison to a simple Linear Regression Model

SUPERGAMING — Project: **Bull Fight** - An online multiplayer strategy based game [April '20 - Jun '20]

- Objective: Creating new points of interaction and content for the players to increase user retention
- Included new **gameplay elements** and implemented a new **First Time User Experience** through **Unity** game engine
- Improved the **core game mechanics** through **C#** scripting and provided new variables that affect the outcome of a game
- Engineered Deck Making mechanism to allow players to customize their deck as per need

Projects Undertaken

C- : C Compiler — Mentor: Prof. Amey Karkare, CSE, IIT Kanpur — Compiler Design [Jan '21 - Apr '21]

- Implemented a **compiler** for **C** programming language with the assembly language generated for **x86_32 bit architecture**
- Used Python library, **PLY** to implement the **lexer** to tokenize the input and **parser** to assign to correct grammatical rule
- Used **3 Address Code** as **intermediate language** to act as bridge between high level source and low level assembly language
- Added support for errors and warnings generation for syntactically and semantically erroneous program

GEM OS — Mentor: Prof. Debadatta Mishra , CSE, IIT Kanpur — Operating System [Aug '20 - Nov '20]

- Implemented various operating system elements on GemOS such as **Paging**, **File system** and **Message Queues**
- Engineered memory management operations with provisions for **mmap**, **munmap** and **huge page creation**
- Created a **debugging system** for GemOS which could be used to get stored variables information at the start of any function.
- Used pipes to synchronize the input and output of programs running in parallel

RESEARCH PORTAL — Mentor: Prof. Sandeep Shukla — Academics and Career Council [May '19 - July '19]

- Spearheaded the creation of an online portal to bridge the gap between students and faculty members
- The portal acts a platform for research related interaction and project collaboration
- Used **Node** for back end, **MongoDB** as database and designed user interface with **HTML5** and **CSS**

POKER AT-7 — Mentor: Prof. Purushottam Kar, CSE, IIT Kanpur — ESC101 [Oct '18 - Nov '18]

- Created an online single-player **POKER** game with the game logic built in vanilla **JavaScript**
- BOT behavior was based on **probability** of different outcomes possible in the current game scenario

C3i LAB, IIT Kanpur — Project: Filtering USB Storage Devices in Linux Based Systems [Dec '19]

- Worked on methods to **allow selective USB Devices** to transfer system data using their unique **Product ID**
- Established **centralized blocking** of devices through configuring rules and updating the rules on all systems

PL BOT — Self Project [Dec '20]

- Created a Discord Bot to provide Premier League fixtures and standings with provisions for sorting the standings on the desired field

Technical Skills

- Languages** : C, C++ , Python, C#, Javascript, NodeJS, Bash Scripting, x86 assembly, HTML5
- Tools** : Github, L^AT_EX, AWS, Pandas, Matplotlib, Numpy, SQL, MongoDB, Unity Engine

Relevant Courses

Data Structures and Algorithms	Software Ops. and Development	Computer Organisation
Discrete Mathematics	Database Management	Multivariable Calculus(A*)
Linear Algebra and ODEs	Advanced Algorithms	Theory of Computation
Machine Learning	Data Mining	Operating System
Compiler Design	Principles of Programming Languages	Computer Networks

A*: Exceptional Performance

Positions of Responsibility

Senator — Students' Gymkhana, IIT Kanpur [Aug '18 - Mar '19]

- A representative **elected by 900 students** to act as their voice and put forth their interests in **Students' Senate**.

Secretary — Programming Club, IIT Kanpur [Apr '19 - Apr '20]

- Conducted events to increase the active participation of students in programming related activities
- Assisted in organizing **competitions, lectures** and **workshops** to achieve the same

Academic Mentor — Counselling Service, IIT Kanpur [Apr '19 - Apr '20]

- **Guided** needy first year students in the course **Fundamentals of Computing** (ESC101)
- Took remedial sessions and mentored students for their academic development and overall upliftment.

Student Guide — Counselling Service, IIT Kanpur [Apr '19 - Apr '20]

- Helped newly joined first year students to get used to campus life and to quickly adjust to campus culture.

Web Executive — Academics and Career Council, IIT Kanpur [Mar '19 - Mar '20]

- **Orchestrated** the creation of council website and was responsible for its maintenance